DATA SCIENCE ALLIANCE • TASK 2.3 CHECK-IN

Downtown San Diego Unsheltered Homeless Population: EDA Findings

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SOURCE DATA

Downtown San Diego Partnership (DSDP) Clean & Safe monthly counts

LONGITUDINAL SCOPE

2017–2025 (8 full & partial calendar years, 7 downtown neighborhoods)

ANALYSIS FOCUS

Overall trajectory, seasonality, COVID dip, methodology & spatial dynamics

Overall Longitudinal Trend: Three Distinct Phases (2017–2025)

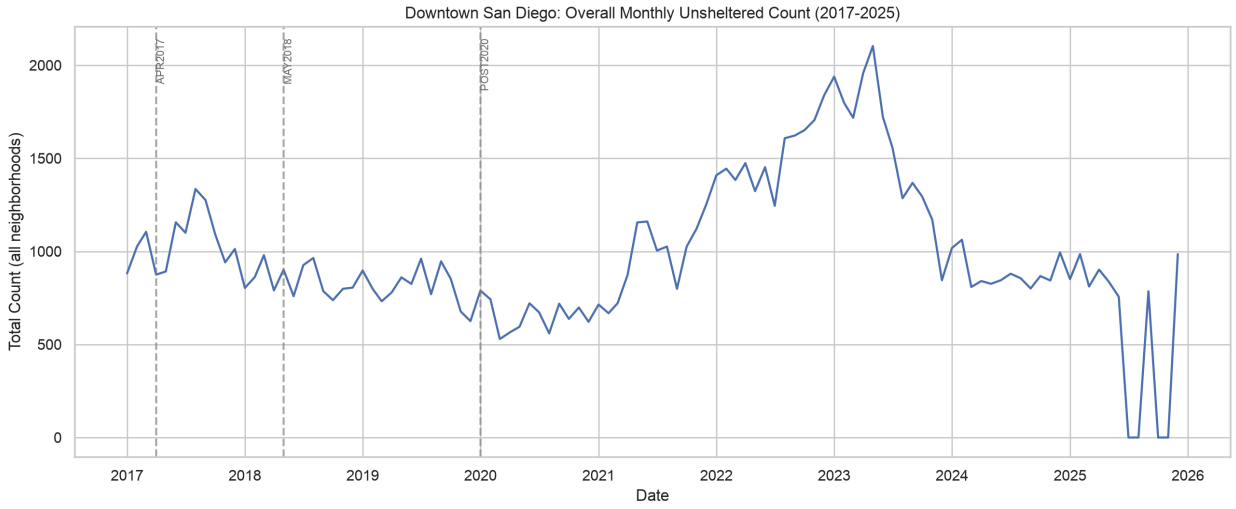
529
SERIES LOW
March 2020

2,103
SERIES PEAK
May 2023

156.4
GRAND MEAN
Per neigh-mo

Trajectory Overview: Three Clear Phases

- **2017–2020 Decline:** Annual average fell from 176.31 to 109.02 (~-38% cumulative drop). YoY changes: -20.31% (2018), -3.87% (2019), and -19.29% (2020).
- **2021–2023 Rebound & Peak:** Counts rebounded sharply: +25.85% (2021) and +57.55% (2022), reaching an annual average of 223.30 in 2023 — more than double the 2020 trough.
- **2024–2025 Pullback & Stabilization:** A steep -43.26% drop in 2024 brought the annual average down to 126.71; 2025 remained roughly flat (-2.56%) at 123.47.



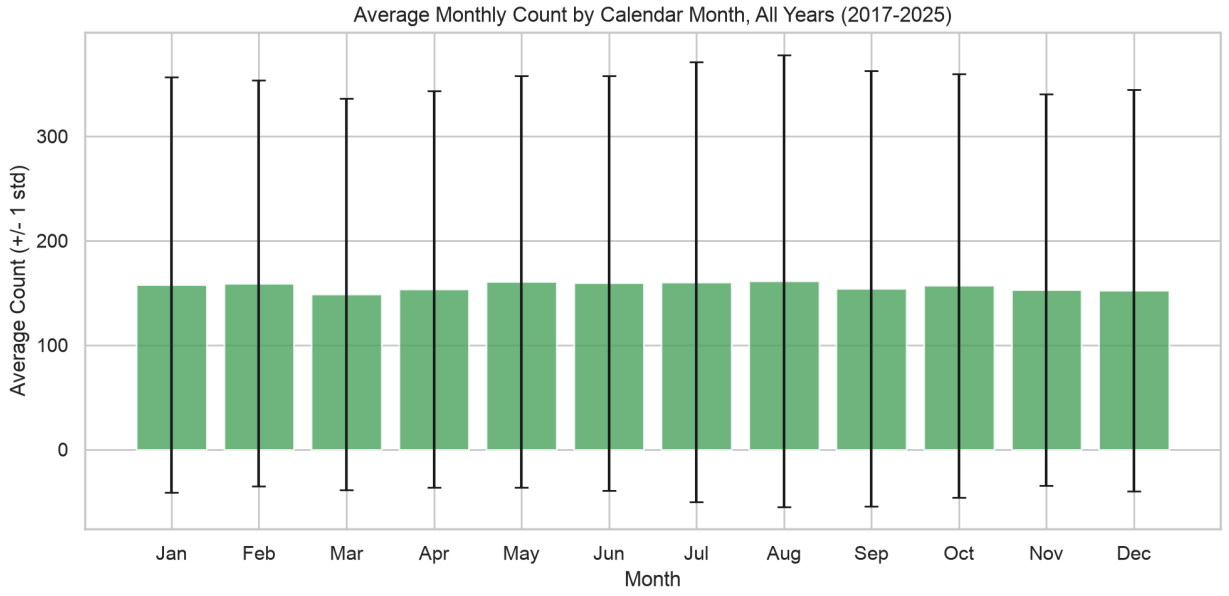
Source: DSDP Clean & Safe monthly counts, MasterCounts_Adjusted_v2.csv

Calendar-Month Seasonality: Modest Variance vs. Annual Swings

Quantifying Calendar Seasonality

Key Takeaway: Which YEAR a month falls in matters far more than which calendar month it is.

- **Modest Range of Deviation:** No single calendar month deviates by more than ~5% from the grand mean (156.44). Peak-to-trough spread is only ~8.1 percentage points.
- **Top 3 Highest Months:** Feb (+3.2%, 161.76), Aug (+3.1%, 161.61), and May (+2.6%, 160.93).
- **Top 3 Lowest Months:** Mar (-5.2%, 148.67), Sep (-2.0%, 153.63), Nov (-2.1%, 153.43) [Apr: -2.1%, 153.53].
- **Large Residual Variance:** Month-to-month standard deviations (187–216) dwarf monthly mean differences, confirming macro annual shocks overpower calendar cycles.
- **East Village Sub-Pattern:** East Village diverges from citywide rhythm: peaks in Jul–Aug (~601–604) and dips in spring (Mar–Apr, ~509).



Source: DSDP Clean & Safe monthly counts, MasterCounts_Adjusted_v2.csv

Neighborhood Rankings: Volume Concentration vs. Growth Rates

Volume Scale vs. Growth Dynamic

Core Insight: Growth is NOT concentrated in the largest neighborhood.

• By Average Monthly Count (Magnitude):

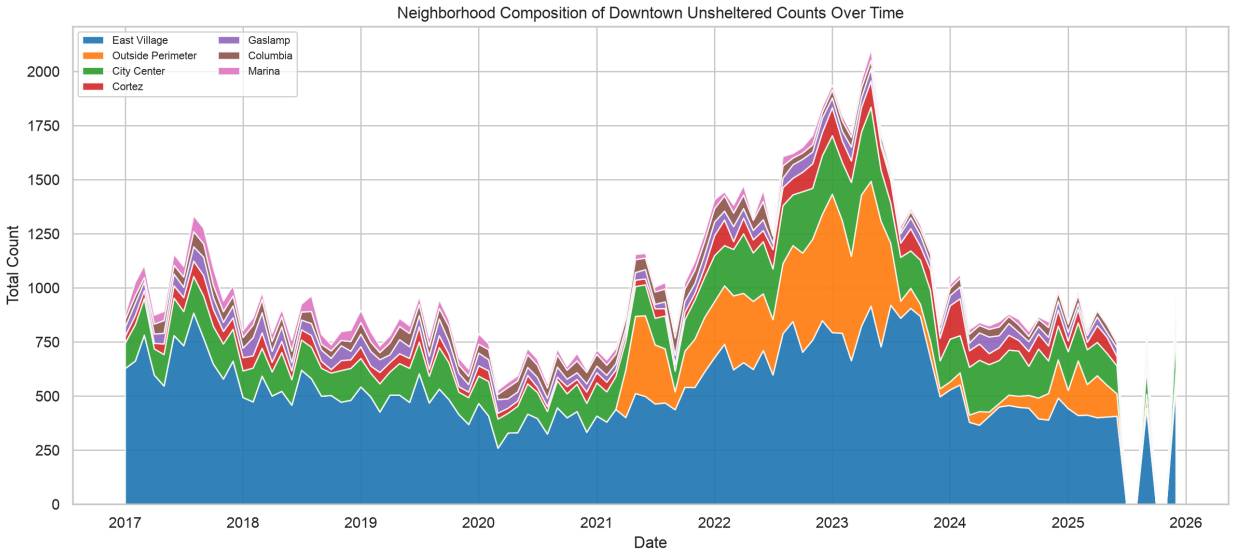
1. East Village: 552.04 (std 157.27 — highest volume & volatility)
2. Outside Perimeter: 217.57 (partial coverage, 56 obs)
3. City Center: 170.68 (std 52.95)
4. Cortez: 59.62 | 5. Gaslamp: 48.17 | 6. Columbia: 44.49 | 7. Marina: 30.75

• By Growth Rate (Mean YoY %, 2019+):

- Fastest Growing: Outside Perimeter (+26.90%/yr), Cortez (+12.12%/yr), City Center (+5.19%/yr).
- Modest Core Growth: East Village (+1.64%/yr) grew slowly despite ~55% total volume.
- Shrinking Areas: Marina (-13.11%/yr) and Columbia (-4.28%/yr) declined.

• Spatial Sub-Area Asymmetry:

Within East Village, North East & South East EV consistently outcount West sub-areas.



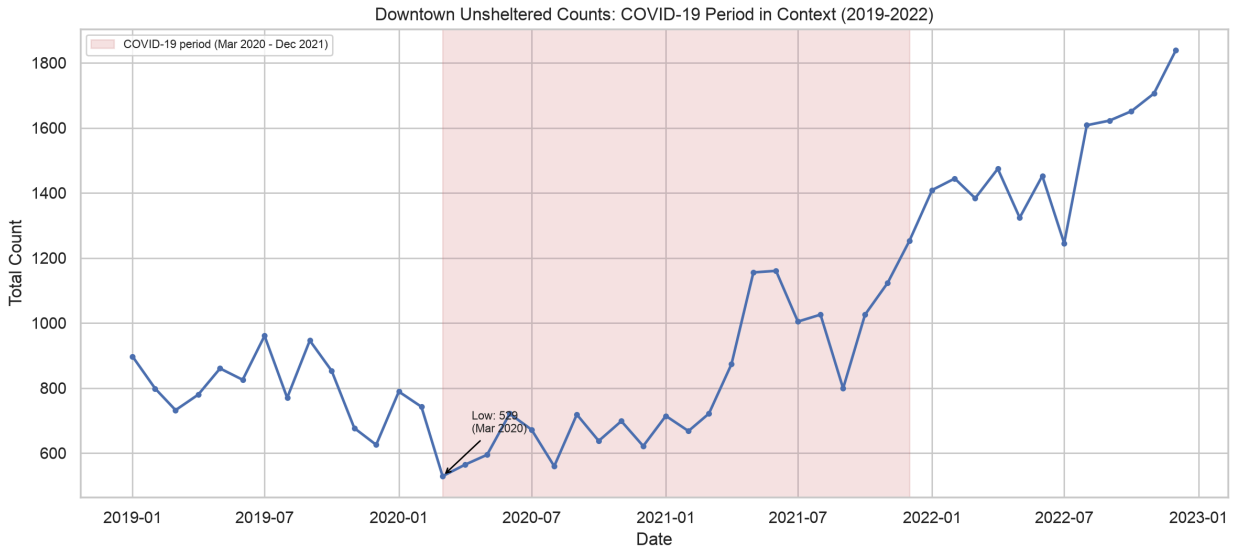
Source: DSDP Clean & Safe monthly counts, MasterCounts_Adjusted_v2.csv

COVID-19 Period Impact: Temporary Suppression & Sharp Rebound

Pandemic Shock & Rebound Profile

Finding: Pandemic programs suppressed visible street counts without a lasting reduction in demand.

- **Series All-Time Low (March 2020):** Citywide unsheltered total hit its series low of 529 in March 2020 at the onset of initial COVID shelter-in-place orders.
- **2020 Annual Trough (~20% Reduction):** 2020 annual average (654.1) dropped ~20% below the 2019 pre-COVID baseline (810.43), reflecting -19.29% YoY contraction.
- **Full Recovery by 2021:** Counts recovered to an annual average of 960.37 in 2021 (+46.82% YoY, citywide-total basis), climbing above pre-COVID baseline levels prior to the 2022–2023 surge.
- **Operational & Policy Mechanisms:** Emergency indoor shelter placements (Convention Center Operation Shelter to Home), eviction moratoria, and curtailed street sweeps temporarily reduced visible unsheltered numbers; demobilization preceded the 2022–2023 peak.



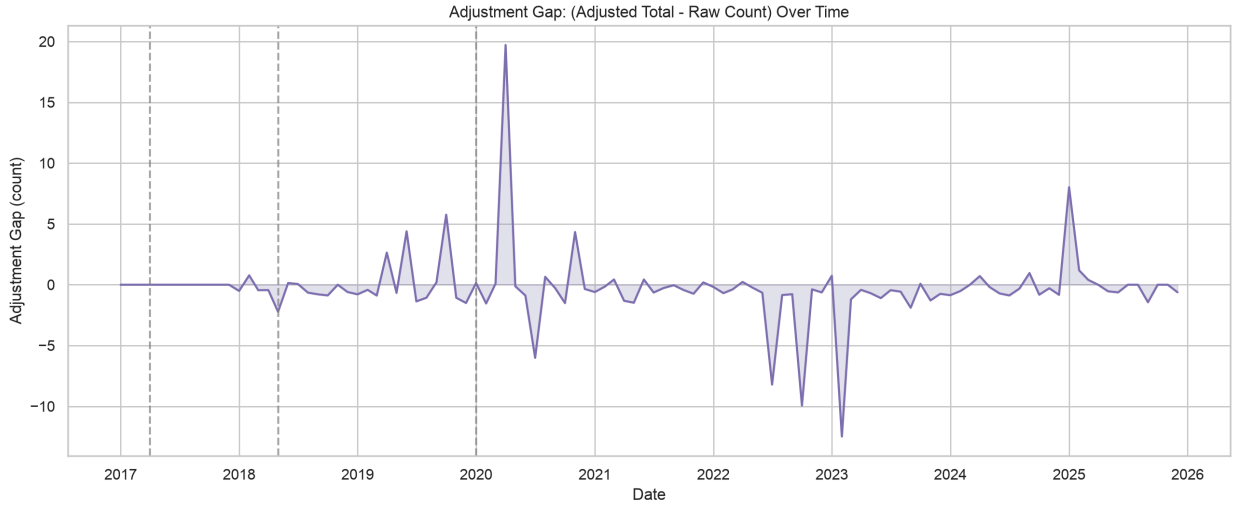
Source: DSDP Clean & Safe monthly counts, MasterCounts_Adjusted_v2.csv

Methodology Regimes: Quantifying the Raw vs. Adjusted Count Gap

Evaluation of Multiplier Regimes

Conclusion: Multipliers add only a small gap (within ±2/month on average) and do not distort longitudinal trends.

- **Four Historical Regimes:**
  - PRE2017 (Jan–Mar 2017): Direct visual count, no component breakout.
  - APR2017 (Apr 2017–Apr 2018): Vehicle multiplier set to 1.66×.
  - MAY2018 (May 2018–Dec 2019): Vehicle multiplier increased to 2.03×.
  - POST2020 (Jan 2020–present): 2.03× retained by DSDP for continuity.
- **Gap Is Small in Every Year:** Gap (total_count - raw_count) averages within ±2/month each year: DSDP published totals already include the multipliers. The earlier Feb 2018 spike (+156.61) was a data-entry error (100 tents instead of 10), corrected against the source map in Oct 2026.
- **Remaining Outliers Are Source Conflicts:** The largest remaining single-month gaps (Apr 2020 +19.7, Feb 2023 -12.5) are months where DSDP map pages and its summary table disagree; listed in MasterCounts_QA_Flags.csv.
- **Pre-2018 Constraint:** Because component breakouts began in Jan 2018, raw and adjusted totals are identical across all of 2017 (gap = 0).



Source: DSDP Clean & Safe monthly counts, MasterCounts_Adjusted_v2.csv

Hypotheses for Further Analysis: Focus Areas for Later Tasks

H1: Policy & Enforcement Driver of 2022–2023 Peak

SYNCHRONIZED CITYWIDE SURGE

The +57.55% (2022) surge hit nearly every neighborhood simultaneously (Cortez +132%, Outside Perimeter +87%, East Village +50%). This synchronized timing points to systemic citywide drivers (eviction moratorium expirations, shelter demobilizations, or sweep shifts) rather than local dynamics.

- **Test: Correlate monthly counts against RTFH policy timelines, shelter census, and 311 encampment dispatch logs.**

H2: East Village Asymmetry Reflects Built Environment

EAST VS. WEST INFRASTRUCTURE DIVIDE

East Village sub-areas show persistent asymmetry: North East and South East EV consistently outcount North West and South West EV across 2018–2025. This concentration indicates infrastructure and zoning drivers.

- **Test: Spatial autocorrelation (Moran's I / Getis-Ord Gi* in Assignment 5) against transit hubs, shelter proximity, and zoning.**

H3: 2024 Pullback (-43.26%) May Be a Reporting Artifact

RAPID SINGLE-YEAR DROP SCRUTINY

A steep -43.26% drop in a single year immediately after the historical peak warrants verification against external benchmarks before reading it as a true population drop. Count area or staffing changes may be confounders.

- **Test: Benchmark against independent RTFH Point-in-Time (PIT) counts and audit 2024 DSDP canvassing operations.**

H4: Neighborhood Growth Inversely Related to Area Size

PERIPHERAL DISPLACEMENT DYNAMICS

Mid-sized areas grow fastest (Outside Perimeter +26.90%, Cortez +12.12%, City Center +5.19%), while East Village grows slowly (+1.75%) and Marina shrinks (-13.11%). Enforcement in core areas may displace individuals to borders.

- **Test: Perform spatial adjacency and boundary-flow analysis in Assignment 5 to evaluate displacement effects.**

Data Limitations, Governance Caveats & Roadmap Ahead

Identified Data Limitations & Biases

- **Incomplete 2025 Series (Reporting Lag):** 28 nulls in total_count (Jul, Aug, Oct, Nov 2025 across all 7 neighborhoods) represent unreleased counts, not data errors. All 2025 annual figures must be treated as partial-year.
- **Outside Perimeter Partial History:** Outside Perimeter contains only 56 observations (canvassing began ~2022). Its reported +26.90%/yr growth rate is the least statistically reliable neighborhood figure.
- **Pre-2018 Lack of Component Breakouts:** Component-level counts (tents, vehicles) are unavailable before Jan 2018. Raw and adjusted totals are identical for all of 2017, precluding early multiplier analysis.
- **Documented Source-PDF Addition Errors:** Task 1.1's 45-cell audit found a 97.56% fidelity rate against source PDFs, noting known errors (Feb 2018 +30 PDF addition bug, 2019 EV sub-area duplication, Apr 2025 discrepancy).

Roadmap Ahead: Assignment 3 Integration

- **Immediate Deliverable: Assignment 3:** Block-Level Validation & Spatial Integration — moving from macro neighborhood aggregates to granular micro-data across 500+ individual block parcels.
- **Granular Block-Level Mapping:** Audit street-level counts, reconcile boundary discrepancies, and align block records with GIS city parcel boundaries.
- **Spatial Autocorrelation Modeling:** Execute Global Moran's I and Local Getis-Ord Gi* hot-spot detection in Assignment 5 to quantify spatial clustering around transit nodes and service hubs.
- **Cross-Dataset Validation & Benchmarking:** Cross-reference DSDP monthly counts with regional RTFH annual Point-in-Time (PIT) benchmarks and City of San Diego 311 service call logs.